

SEIRify: A user-friendly R-Shiny application for post-outbreak analysis of community-level infectious disease data

Introduction

Community-level infectious disease outbreaks pose significant challenges to public health systems, where early detection and response are critical to limiting transmission and reducing disease burden (1, 2). Frontline health workers, epidemiologists, public health practitioners, and researchers rely on accurate data analysis tools to understand outbreak dynamics, evaluate control strategies, and inform future preparedness efforts. However, many existing analytical platforms (3) and general-purpose statistical software like R (4) and Python (5) modeling packages require advanced programming skills or statistical expertise. These technical demands make it difficult for rapid and widespread use during and after outbreaks, especially among frontline health workers who may not always have specialized training in mathematical modelling of infectious diseases.

Mathematical models of disease transmission, particularly the Susceptible–Exposed–Infectious–Recovered (SEIR) framework, have long been central to understanding infectious disease transmission dynamics, projecting outbreak trajectories, and evaluating the impact of interventions (6,7,8). These models conceptualize the spread of infectious agents within single or multiple host populations using mathematical language. Through key epidemiological parameters such as the basic reproduction number (R_0), incubation period, and serial interval, insights into the factors influencing disease dynamics can be identified.

However, integrating SEIR modeling into practical, user-friendly tools remains limited, especially in settings where local health workers and researchers may not have extensive computational backgrounds (9). There is a growing need for intuitive and accessible tools that enable retrospective evaluation and post-outbreak analysis without requiring in-depth modeling expertise (10).

To address this gap, we present SEIRify, a simple, interactive, and user-friendly R Shiny application designed for post-outbreak analysis of community-level infectious disease data. SEIRify allows users to fit SEIR-based models, visualize the disease trajectories in compartments, and generate estimates of critical model parameters through an accessible web interface. SEIRify

provides two distinct modes of analysis. In the simulated scenario mode, users define epidemiological parameters (initial population conditions, transmission, exposure, and recovery rates) to generate epidemic trajectories under controlled assumptions. This mode is designed for exploratory modeling, allowing users to test “what-if” scenarios, such as varying R_0 , altering the timing of interventions, or simulating different population sizes. In the comparison mode, SEIRify overlays simulated epidemic curves with real-world outbreak data uploaded by the user. This enables direct visual and numerical assessment of model performance, facilitating evaluation of parameter plausibility and model fit.

We introduce the SEIRify application, outline its technical framework, and demonstrate its functionality. We provide a practical, open-source tool to support post-outbreak evaluation, capacity building, and preparedness for future epidemics.

1.0 Materials and Methods

To launch the application, users must first open [SEIRify](#) in any browser of their choice. While [SEIRify](#) is hosted within the RStudio integrated development environment (IDE), no previous experience with R or RStudio is required to use this application.

After the user interface has been successfully loaded, the dashboard will appear. [SEIRify](#) has 2 main pages: 1) About, and 2) Explore models

About page

Upon initially loading the application, users will see the “About” icon at the top right side of the application. This page provides an overview of the methods employed within the dashboard, including model specification, forecasting, and details of the available evaluation statistics.

Explore models page

The *Explore models* page serves as the starting point for using the [SEIRify](#) dashboard (Figure 1), including specifying the scenario of interest (Simulated Scenario Only versus Simulated and Real Scenario compared), compartmental model type (SI, SIR, and SEIR), inputting disease parameters based on selected models (transmission rate, exposure rate and recovery rate) and inputting the population parameters (Total population, initial susceptible, exposed, infected and recovered).

After selecting all necessary options and clicking the “Run model simulation” button, the resulting models fit and display the appropriate model visualizations and model parameters on this page. Note that each time any input is changed, the “Run model simulation” button has to be clicked again. Note that if options have not been specified and processed, the dashboard will remain blank (Figure 1). The model parameter window displays the key inputs and estimated parameters used in the simulation, allowing users to verify the correctness of the model setup.

The Comparison window provides model outputs of the real versus simulated scenarios. These include key epidemiological metrics, such as the reduction in total infections, the reduction in outbreak duration (in days), the reduction in peak size, the delayed epidemic start (in days), and the delay in the timing of the peak (in days). For each metric, the window displays the point estimate along with its 95% confidence interval (CI).

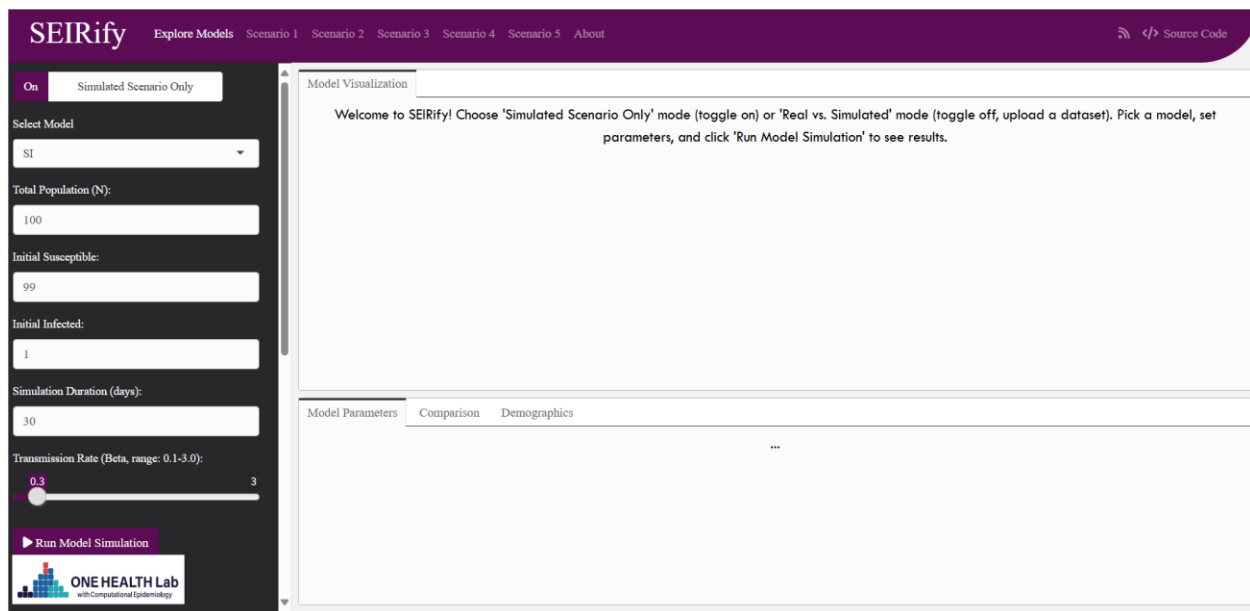
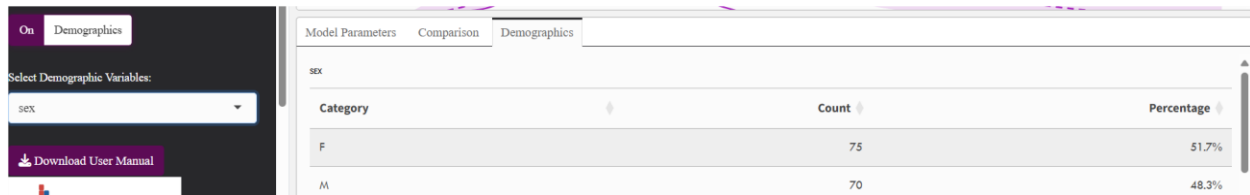


Figure 1. A screenshot of the *Explore models* page after initially loading the dashboard.

Users primarily interact with the dashboard’s sidebars, where details related to the population, scenario specifications, and all model parameters must be entered. Once selected, the user can then click “Run model simulation” to obtain the output. The [SEIRify](#) dashboard was developed using R-Shiny.

The Demographics window shows a summary of all the variables of the uploaded dataset. Numeric variables are summarized as means and standard deviations, while categorical variables are presented as counts and percentages.



Category	Count	Percentage
F	75	51.7%
M	70	48.3%

Figure 2. A screenshot of the Demographics tab

The scenarios

SEIRify allows users to explore two types of scenarios, the simulated scenario and the real versus simulated scenario. To activate each scenario, the toggle must be switched on for the simulated scenario and off for the real versus simulated scenario (Figure 2). In the simulated scenario, the model generates plots and model parameters for each compartment based only on user-defined parameters, including population size, transmission rate, recovery rate, and simulation duration. This mode is useful for examining hypothetical situations. In the real versus simulated scenario, users are expected to upload an outbreak dataset, and the model compares simulated epidemic curves with the observed data.



Figure 3. A screenshot showing how to activate both the simulated scenario only and the real versus simulated scenario

The input data

In SEIRify, users can upload their own outbreak datasets in CSV format using the “Choose CSV File” option for the ‘real versus simulated scenario’ ONLY. For users who do not have a dataset, a mock dataset can be downloaded directly from the interface at the bottom left corner (‘Download Mock Data’). Note that this is a dummy dataset for demonstration. The input data

must follow a structured format with specific variables. The uploaded dataset must have the following required variables and format;

- ✓ date_of_exposure – the date when an individual was exposed to infection in YYYY-MM-DD format.
- ✓ date_of_onset – the date when symptoms or infection onset occurred in YYYY-MM-DD format.
- ✓ Any other demographic or disease-related required variable (Figure 3).

1	id	date_of_exposure	regions	date_of_onset	Source of infection	type	country	Transmission type	Age group	sex
2	#1	2021-11-24	incheon	2021-11-25	11.24 Arrivals from Nigeria(ET672)	abroad	Nigeria	Imported	40s	M
3	#2	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	40s	F
4	#3	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	40s	M
5	#4	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	30s	F
6	#5	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	50s	M
7	#6	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40s	F
8	#7	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	60s	M
9	#8	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	30s	M
10	#9	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40s	F
11	#10	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40s	F
12	#11	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	30s	F
13	#12	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	50s	F
14	#13	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40s	M
15	#14	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40s	M
16	#15	2021-11-25	Daegu	2021-11-26	Patient #1	National	Domestic	Domestic	30s	F
17	#16	2021-11-26	Daegu	2021-11-26	Patient #15	National	Domestic	Domestic	40s	M
18	#17	2021-11-25	Daegu	2021-11-27	Patient #1	National	Domestic	Domestic	30s	F

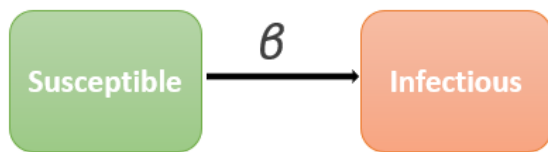
Figure 4. Screenshot of the data CSV file

The models

The current version of SEIRify provides users with the opportunity to model three different compartmental models (Figure 4):

- ✓ Susceptible (S) – Infectious (I): SI Model

In SI models, individuals who become infected remain infectious for life, as infection does not confer immunity. Thus, people never leave the infectious state. This structure is suitable for modeling lifelong infections such as HIV or herpes virus. The key parameter is the transmission rate (β), which represents the probability of disease transmission between a susceptible and an infectious individual per contact per unit time.



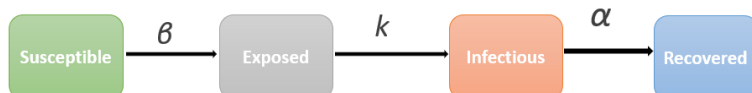
✓ Susceptible (S) – Infectious (I) – Recovered (R): SIR Model

In SIR models, infected individuals eventually recover and move into the recovered class, where they are assumed to be immune. The two main parameters are the transmission rate (β) and the recovery rate (α), with α defined as the reciprocal of the mean infectious period (1/mean infectious period).



✓ Susceptible (S) – Exposed (E) – Infectious (I) – Recovered (R): SEIR Model

The SEIR model adds an exposed class to capture the incubation period during which individuals are infected but not yet infectious. This model is widely used for acute infectious diseases with a latent stage. Parameters required for this model include the transmission rate (β), the progression rate (k) from exposed to infectious (defined as the reciprocal of the incubation period, 1/incubation period), and the recovery rate (α) (1/mean infectious period).



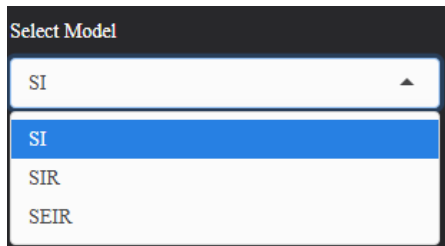


Figure 5. Screenshot displaying the compartmental models in SEIRify

Depending on the user's interest, SEIRify provides the option to visualize the dynamics of individual model compartments separately. By selecting “Show compartments separately”, users can isolate and plot the trajectory of a chosen compartment (e.g., susceptible, exposed, infected, or recovered). In addition, the interface allows the display of confidence intervals for either the simulated scenario, the real scenario, or both, with selectable levels (95%, 90%, 75%, or 50%). This functionality enables more detailed exploration of compartment-specific dynamics and facilitates comparison between real and simulated outcomes.

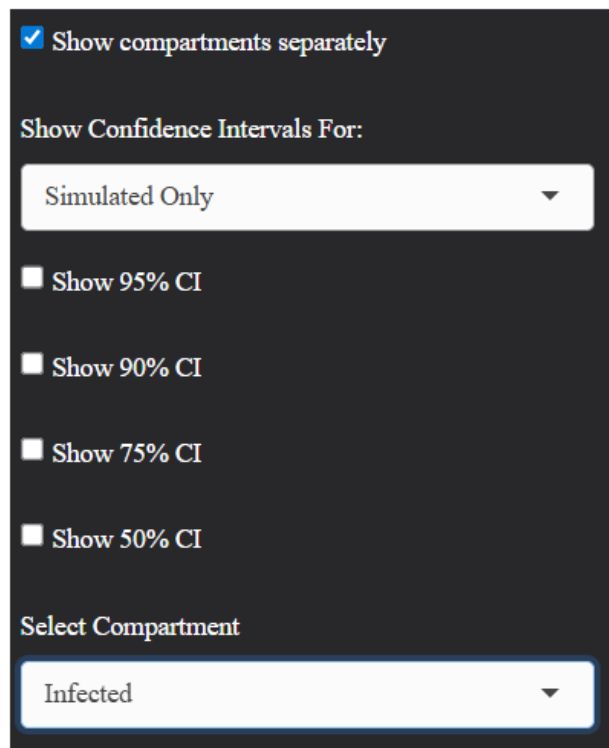
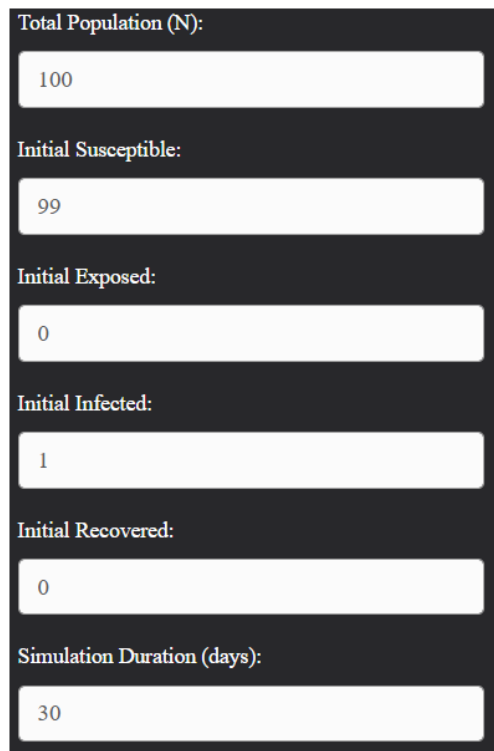


Figure 6. Screenshot showing selection of the ‘show compartment separately’ icon and its specificities

The population

In SEIRify, users are required to specify the total population (N) for which the simulation will be performed. Once this value is entered, the system automatically adjusts the counts for the initial susceptible and initial infected populations according to the user's input. This setup reflects the epidemiological principle that an outbreak typically begins with an index case and subsequently spreads to susceptible individuals in the community. Users must also provide the simulation duration (days), which determines the time horizon of the epidemic projection in the simulated scenario. For the real versus simulated scenario, however, the epidemic duration is determined by the range of dates present in the uploaded dataset. An additional optional input allows users to define the total population size in the real versus simulated scenario, enabling estimation of the uninfected population as the difference between the total community population and the number of infected individuals recorded in the uploaded dataset.



The screenshot shows a dark-themed interface with several input fields. Each field has a label in a lighter color and a corresponding input box. The labels and their values are: 'Total Population (N):' with '100', 'Initial Susceptible:' with '99', 'Initial Exposed:' with '0', 'Initial Infected:' with '1', 'Initial Recovered:' with '0', and 'Simulation Duration (days):' with '30'.

Parameter	Value
Total Population (N)	100
Initial Susceptible	99
Initial Exposed	0
Initial Infected	1
Initial Recovered	0
Simulation Duration (days)	30

Figure 7. A screenshot of the population input fields showing the initial conditions and simulation duration for the outbreak model

The epidemiological parameters

Based on the selected model, users are required to specify the appropriate epidemiological parameters. These include the transmission rate (β), defined as the probability of infection per

contact per unit time; the progression rate (k) from exposed to infectious, which is the reciprocal of the incubation period ($1/\text{incubation period}$); and the recovery rate (α), defined as the reciprocal of the mean infectious period ($1/\text{mean infectious period}$). In SEIRify, these parameters are adjusted interactively by moving the corresponding toggles on the interface, allowing users to explore different epidemic trajectories under varying assumptions of disease transmissibility, latency, and recovery.

Output

SEIRify provides multiple options for accessing and exporting model outputs. Visual results are displayed in the Model Visualization window, where epidemic dynamics can be plotted by compartment for both simulated and real scenarios. These plots can be downloaded directly as PNG images or PDF documents for reporting and presentations. In addition, the underlying model parameters are displayed in the Model Parameters window, with the option to export the data in CSV, Excel, or PDF formats. Similarly, the Comparison window summarizes key performance metrics between simulated and real scenarios (reduced duration, delayed start, reduced infections, reduced peak, or delayed peak), which can also be downloaded in the same formats.

Application specifications

To ensure broad accessibility, SEIRify was developed as a web-based application that can be accessed through a standard internet browser without the need for specialized software installation. The framework was built on RStudio Shiny. The epidemic models were implemented as R scripts, with each model encapsulated in a dedicated R function and invoked by the server-side script (server.R) of the Shiny application. This design maintained a clear separation between the user interface and computational components, allowing model code to be modified independently of the application interface. The SEIRify application was developed using several packages to support its functionality. The shiny, shinyWidgets, and shinyjs packages provide the core framework for building the interactive dashboard and extending functionality. flexdashboard was used to structure the user interface, while highcharter supported interactive visualizations, complementing the built-in plotting features. For epidemiological modeling, deSolve was employed to solve systems of ordinary differential equations, and EpiEstim was used for estimating time-varying reproduction numbers. Data cleaning and management were facilitated by tidyverse, tidyr, data.table, and janitor, with DT enabling interactive tabular displays. lubridate

was used for handling dates and times, which are central to epidemic datasets. Font management and customization of visual outputs were achieved through `extrafont`, `extrafontdb`, `systemfonts`, and `sysfonts`.

Below are the epidemiological parameters for some selected infectious diseases. These parameters can be used for both the simulated scenario only and the real versus simulated scenario.

Parameter	Infectious Disease					
	COVID-19	Influenza A	Measles	Pertussis	Ebola	mpox
Transmission rate per day- β	Wild 0.21 – 0.28 Omicron 1.14 – 2.0	1.8	1.7 – 2.6	0.2-0.5	0.2 to 0.3	0.01 to 0.1
Exposure rate – σ (1/Incubation Period)	0.21 (1 / 4.6 days)	0.5 (1/2 days)	0.08 (1/13 days)	1/7 – 1/10	1/2 – 1/21	1/10
Recovery rate – γ (1/duration of infectiousness)	0.1 – 0.2 ($\approx$ 1 / 5 – 10 days)	0.2 (1/5days)	0.1 – 0.14 (1/7 – 10 days)	0.071	1/6 – 1/16	0.05

Case study 1: Simulated Scenario ONLY - Influenza A

To demonstrate the simulated functionality of SEIRify, we modeled an influenza A outbreak in a hypothetical community of 100 individuals over a 30-day period using the SEIR compartmental model. At the start of the simulation, the population was initialized with 99 susceptible, 1 infected, 0 exposed, and 0 recovered individuals, reflecting the introduction of a single index case into an otherwise susceptible population.

On

Simulated Scenario Only

Select Model

SEIR

Total Population (N):

100

Initial Susceptible:

99

Initial Exposed:

0

Initial Infected:

1

Initial Recovered:

0

Simulation Duration (days):

30

Figure 8. Screenshot showing initial input of population parameters

The model parameters were specified as follows: transmission rate (β) of 1.8 per contact per unit time, exposure rate (σ) of 0.5 (corresponding to an average incubation period of 2 days), and recovery rate (γ) of 0.24 (corresponding to an average infectious period of approximately 4 days). These parameter values represent plausible ranges for influenza A transmission dynamics as published in empirical literature.

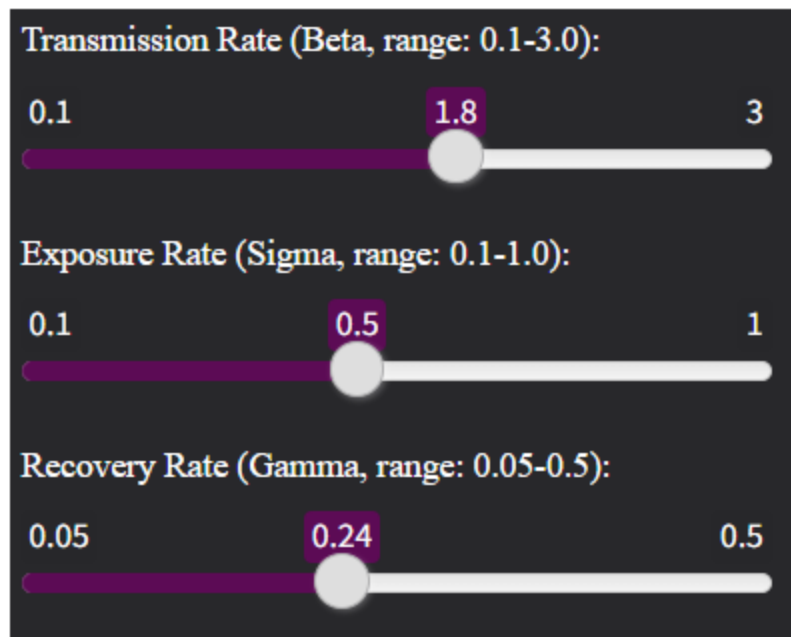


Figure 9. Screenshot showing initial input of transmission parameters

The outputs

Visualization

After selecting the “*show compartment separately*” option, we visualized each compartment individually to better understand the dynamics of the simulated outbreak. Visual plots for each compartment downloaded from SEIRify are shown below.

The susceptible (S) compartment

The susceptible population declined rapidly over the 30-day simulation period. Beginning with 99 susceptible individuals at the onset of the epidemic, the curve shows an accelerated decrease between days 5 and 10 as exposure and transmission intensified. By approximately day 12, the number of susceptible individuals dropped close to zero, indicating that nearly the entire population had transitioned to exposed or infected compartments.

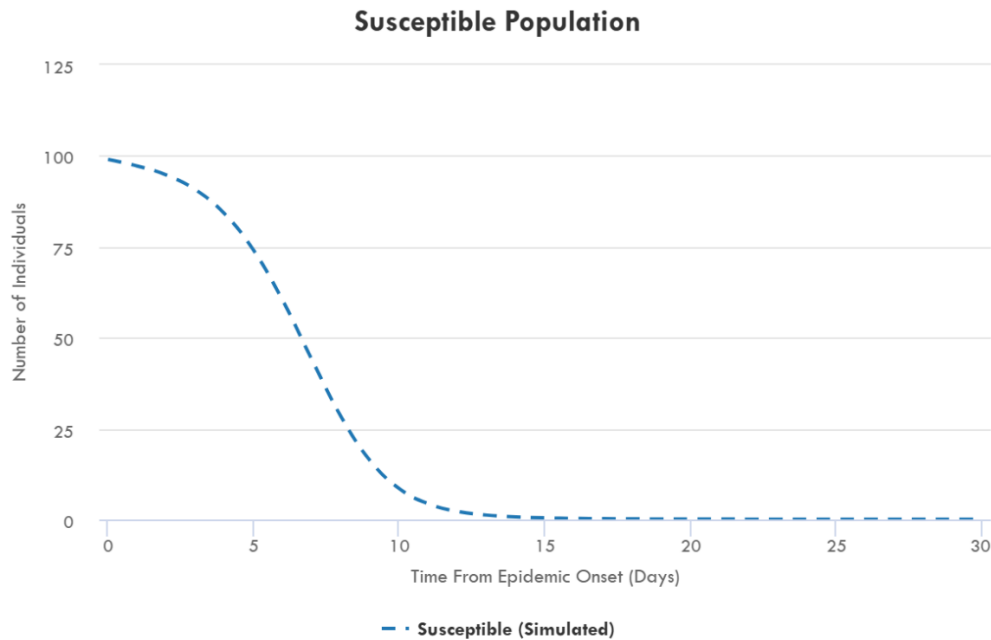


Figure 10. Output for the simulated scenario S compartment

The Exposed (E) compartment

The exposed population followed a bell-shaped trajectory during the 30-day simulation. Starting at zero, the number of exposed individuals rose steadily as susceptibles became infected but had not yet developed symptoms. The curve peaked around day 8, with approximately 28 individuals in the exposed state, reflecting the maximum buildup of latent infections. After the peak, the exposed population declined sharply as individuals transitioned into the infectious compartment. By day 20, the exposed compartment had nearly returned to zero, indicating depletion of the latent pool as the epidemic progressed.

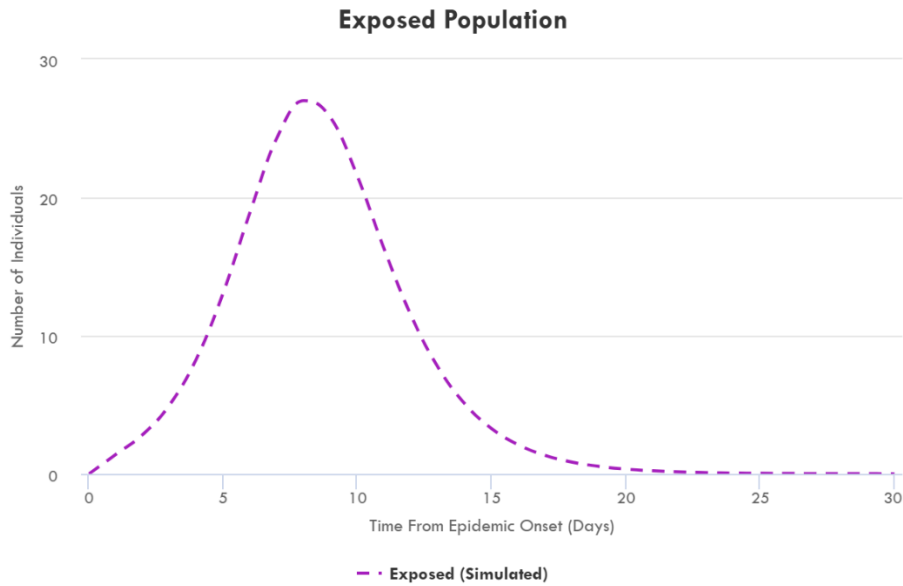


Figure 11. Output for the simulated scenario E compartment

The Infected (I) compartment

Beginning with a single index case, the number of infected individuals increased gradually in the first few days, followed by a sharp rise as transmission accelerated. The curve peaked around day 10, with nearly 38 individuals simultaneously infectious, representing the maximum burden of active cases in the community. After this peak, the number of infected individuals declined steadily as recoveries outpaced new infections.

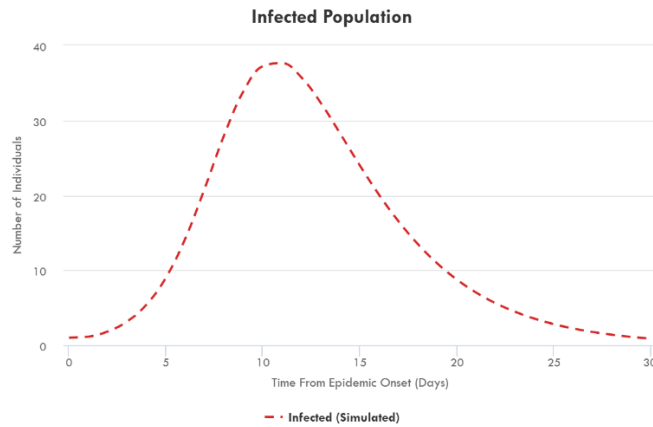


Figure 12. Output for the simulated scenario I compartment

The Recovered (R) compartment

The recovered population increased gradually during the early days of the simulation, reflecting the time lag between infection and recovery. From day 7 onward, recoveries accelerated as more individuals transitioned out of the infectious state. The curve rose steadily until around day 20, when it began to plateau as the epidemic waned and the susceptible population was nearly exhausted. By the end of the 30 days, the entire community had entered the recovered compartment, highlighting the self-limiting nature of the outbreak under the assumed SEIR model parameters.

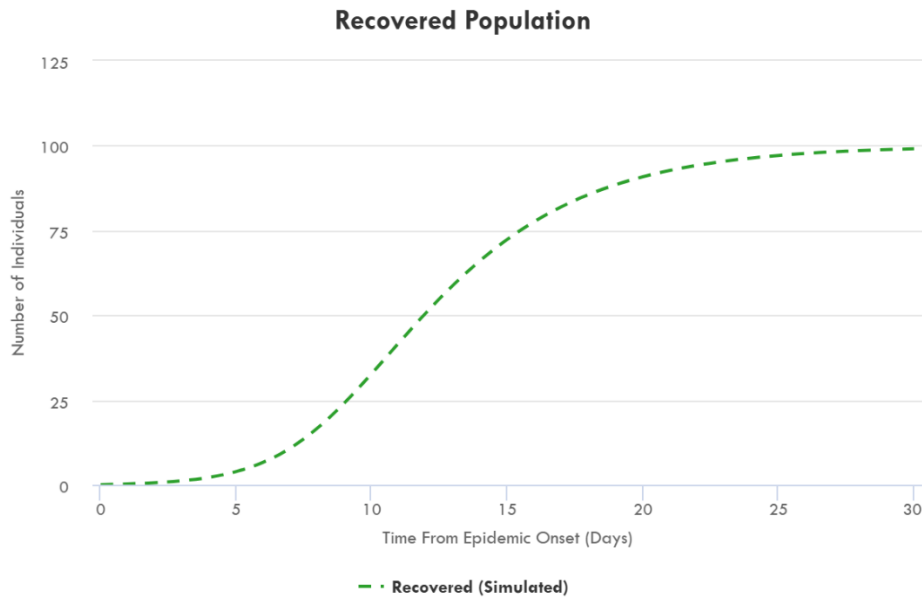


Figure 13. Output for the simulated scenario R compartment

Model parameters

The model parameters downloaded from SEIRify are shown in Table 1. The output showed that the epidemic reached its peak on day 11, when 38 individuals were simultaneously infected.

Table 1. Simulated only model parameters

Parameter	Description	Value
N	Total population size	100
Beta	Transmission rate (range: 0.1-3.0 per day)	1.8
Sigma	Exposure rate (range: 0.1-1.0 per day)	0.5
Gamma	Recovery rate (range: 0.05-0.5 per day)	0.24
R0	Basic reproduction number (Beta/Gamma)	7.5
Initial S	Initial susceptible population	99
Initial E	Initial exposed population	0
Initial I	Initial infected population	1
Initial R	Initial recovered population	0
Final S	Final susceptible population	0

Final E	Final exposed population	0
Final I	Final infected population	1
Final R	Final recovered population	99
Duration (days)	Duration of epidemic	30
Peak Infections	Peak number of infected	38
Peak Time	Day of peak infection	Day 11
Incubation Period	Average incubation period (1/Sigma)	2
Infectious Period	Average infectious period (1/Gamma)	4.2

Case study 2: Real versus Simulated Scenario with COVID-19

To demonstrate the real versus simulated functionality of SEIRify, we used a linelist of de-identified COVID-19 data. The dataset contains 10 variables: *ID*, *date_of_onset*, *date_of_exposure*, *regions*, *Vaccine_status*, *type*, *country*, *Transmission.type*, *Age.group*, and *sex*. This tutorial dataset is publicly available at

To begin the analysis, the “*Simulated Scenario Only*” option was switched off, enabling the real versus simulated scenario mode. The de-identified dataset was then uploaded, and the model of interest was selected for comparison. For this example, a total population of 330 was specified to represent the size of the community under study, allowing estimation of both infected and uninfected individuals during the course of the epidemic.

Simulated Scenario Only ☐ Off

Choose CSV File

Browse... mock_data-2025-07-23.csv

Upload complete

Select Model

SEIR

☒ Optional: Input Total Population

Total Population (N)

10 330 1,000

10 109 208 307 406 505 604 703 802 901 1,000

Figure 14. A screenshot of the model setup page showing the real versus simulated scenario mode, dataset upload, SEIR model selection, and specification of the total population size.

The model parameters were specified as follows: a transmission rate (β) of 2.0 per contact per unit time, an exposure rate (σ) of 0.21, corresponding to an average incubation period of approximately 5 days, and a recovery rate (γ) of 0.20, corresponding to an average infectious period of 5 days. These parameter values fall within the plausible ranges for COVID-19 transmission dynamics as reported in the empirical literature.

Transmission Rate (Beta, range: 0.1-3.0):

0.1 2 3

Exposure Rate (Sigma, range: 0.1-1.0):

0.1 0.21 1

Recovery Rate (Gamma, range: 0.05-0.5):

0.05 0.2 0.5

Figure 15. Screenshot of the real versus simulated model parameters

The Output

Visualizations

After selecting the “*show compartment separately*” option, we visualized each compartment for both the real and simulated scenarios side by side. To focus on the stability of the model estimates, we selected the confidence intervals for the simulated scenario only. Visual plots for each compartment, downloaded directly from SEIRify, are presented below.

The Susceptible (S) compartment

The susceptible population declined over time in both the simulated and real scenarios. In the simulated scenario, the susceptible curve showed a rapid decline, with most individuals transitioning out of susceptibility by day 15. However, the real data reflected a more gradual decline, with a sizeable number of individuals remaining susceptible beyond day 30.

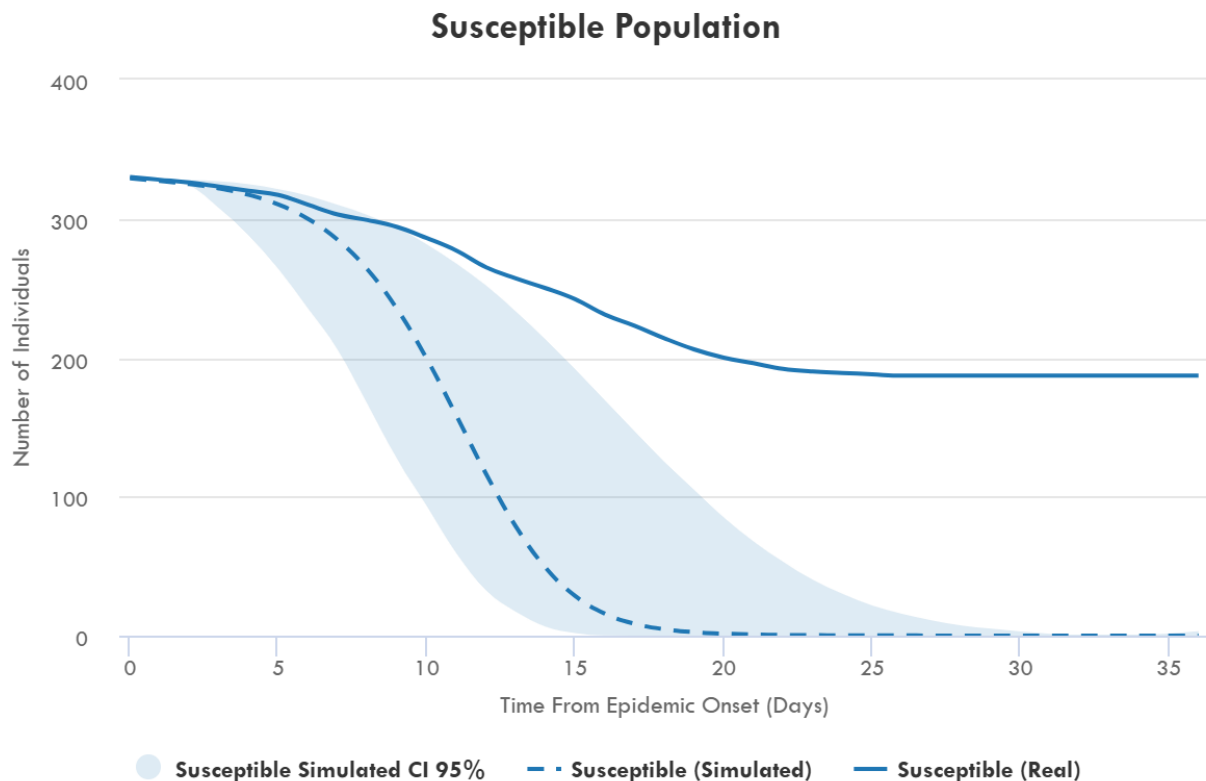


Figure 16. Output for the real versus simulated scenario S compartment

The Exposed (E) compartment

The exposed population showed distinct patterns between the simulated and real data scenarios. In the simulated outbreak, the number of exposed individuals rose sharply, peaking around day 12 with approximately 140 cases before declining. In contrast, the real data reflected a much smaller and flatter exposed curve, with the maximum number of exposed individuals remaining below 30 and distributed over a longer period.

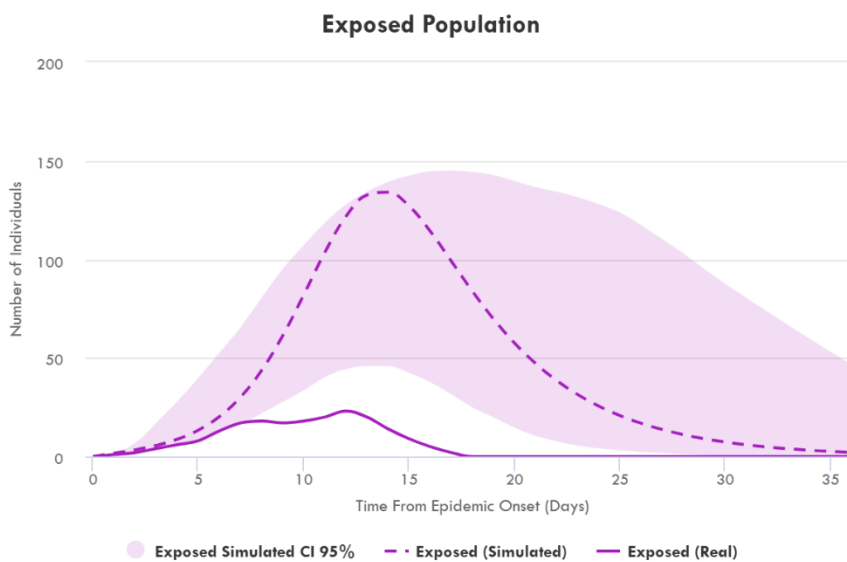


Figure 17. Output for the real versus simulated scenario E compartment

The Infected (I) compartment

The infected population displayed the characteristic epidemic curve in both the simulated and real scenarios, though with differences in peak intensity and timing. In the simulated outbreak, infections rose rapidly, reaching a peak of just over 100 individuals around day 17 before gradually declining. The 95% confidence interval captured the range of plausible trajectories but consistently indicated a sharp and high peak. In the real data, however, the epidemic peak was lower, with approximately 60 individuals infected simultaneously, and the curve was broader, indicating a slower rise and fall in cases. This could suggest that while the simulation predicted a more intense

and concentrated epidemic wave, the real dataset reflected a less severe but more prolonged transmission dynamic.

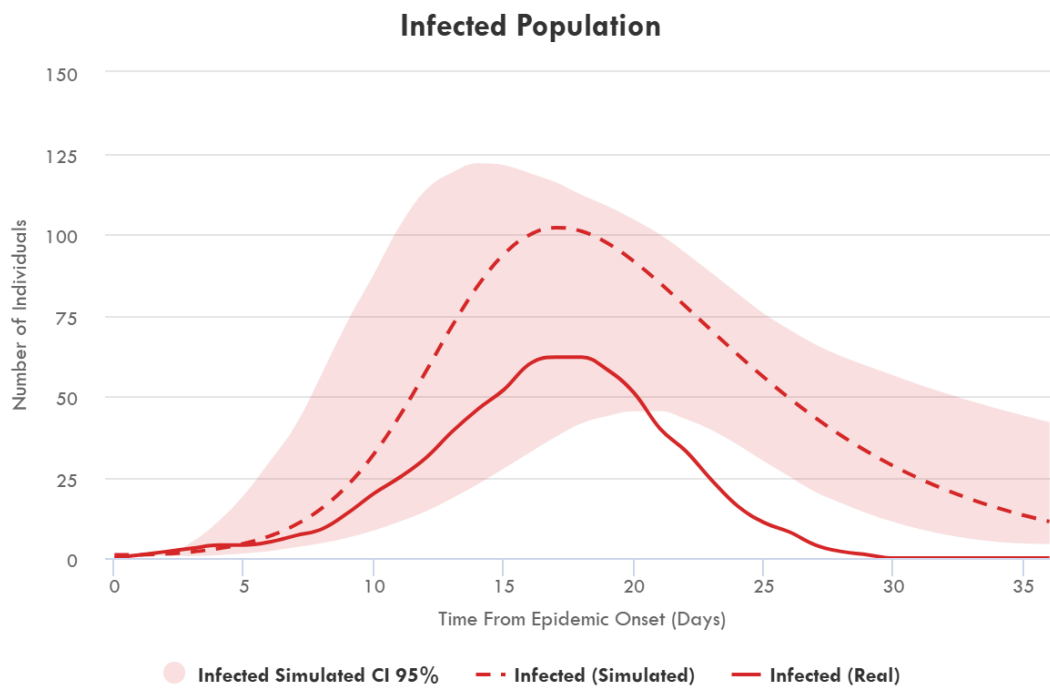


Figure 18. Output for the real versus simulated scenario I compartment

The Recovered (R) compartment

The recovered population increased steadily in both the simulated and real scenarios but differed in timing and magnitude. In the simulated outbreak, recoveries began to rise sharply around day

10, while the real data showed a slower and more gradual accumulation of recoveries.

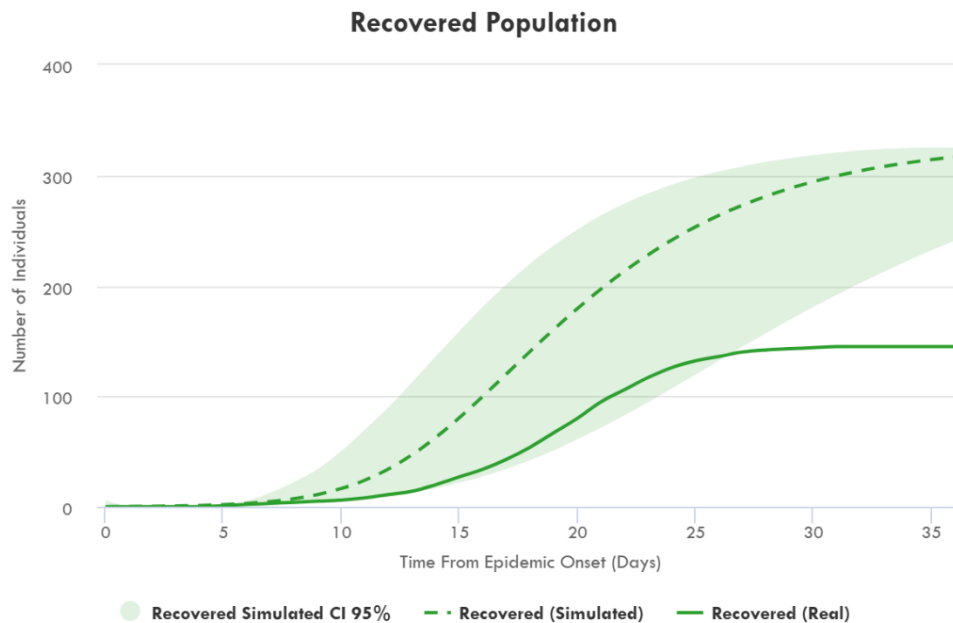


Figure 19. Output for the real versus simulated scenario R compartment

Comparison between the simulated and real scenarios

The comparison of real versus simulated scenarios produced the following epidemic metrics. The simulation predicted 4 more infections than were observed in the real dataset (Estimate: -4 , 95% CI: -6 to -4). There was no difference in epidemic duration, with both scenarios spanning a similar number of days as expected. However, the peak number of infections was estimated to be 21 cases higher in the simulation scenario (95% CI: 13–28), suggesting the model projected a more intense epidemic wave. The timing of the epidemic peak was delayed by 2 days in the simulated scenario compared to the real data (95% CI: -1 to 3), while the epidemic onset was also delayed by 2 days (95% CI: 1–4).

Table 2. Out of the comparison window of the real versus simulated scenario

Metric	Estimate	Lower CI	Upper CI
Reduced Infections	-4	-6	-4
Reduced Duration (days)	0	0	0
Reduced Peak	21	13	28

Delayed Peak (days)	2	-1	3
Delayed Start (days)	2	1	4

Technical validation

SEIRify was validated through a series of simulation checks and exercises. Model outputs from the SI, SIR, and SEIR implementations were compared against independently coded models, with consistent epidemic trajectories observed. Validation with mock outbreak datasets confirmed that simulated versus real scenario comparisons accurately reproduced epidemic metrics such as peak timing, total infections, and outbreak duration. Output remained stable across different computing environments (local RStudio, web-based, and offline use), ensuring reproducibility. Interface stress-testing demonstrated that parameter adjustments and dataset uploads were correctly propagated to visualizations and downloadable outputs.

Conclusion

Post-outbreak community infectious disease modeling and forecasting provide an understanding of epidemic dynamics, direction for intervention planning, and policy decision-making. As shown above, SEIRify provides an interactive interface for modeling post-outbreak diseases in the community. The application supports both simulated scenarios and real versus simulated comparisons, enables visualization of epidemic dynamics across compartments, and allows for customizable parameter inputs, while offering outputs in multiple formats. Its strength lies in providing a user-friendly, programming-free environment for retrospective epidemic modeling, making modeling techniques more widely accessible to health workers with no modeling background.

Although SEIRify broadens access to compartmental modeling, it is not without limitations. First, the platform assumes users possess a basic understanding of infectious disease dynamics and the interpretation of epidemiological parameters. Second, it does not provide detailed modeling guidance beyond short descriptions of input variables and model outputs.

Overall, SEIRify, offers a programming-free, interactive tool for visualizing epidemic dynamics, facilitating side-by-side comparisons of real and simulated outbreaks, and fostering greater engagement with infectious disease modeling among professionals.

Availability and requirements

Application name: SEIRify

Application home page: <https://achangwa.shinyapps.io/SEIRify/>

Operating system: Platform independent.

Programming language: R.

Other requirements: Any web browser,

Any restrictions on use: Organizations are welcome to contact the author before use.

Conflict of interest

We declare that we have no commercial or financial relationships that could be perceived as a potential conflict of interest in the development of this application.

Acknowledgement

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